



## CONTACT INFO

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## EDUCATION

### BSc (Hons) Computer Science

Univ. of Bedfordshire (Top-up)

*Planned Specialization*

### Higher Diploma in IT

SLIIT City Uni

*Final Semester (Current)*

### Foundation Cert. in IT

SLIIT City Uni

*Completed*

### G.C.E. Ordinary Level

St. Joseph Vaz College

**Achievement: 9 'A' Distinctions**

## CERTIFICATIONS

### Certificate in AI

SLIIT City Uni

*Machine Learning & Models*

### Diploma in English

BCI Campus

*Professional Communication*

# L.B. LOCHANA CHAMOD

## COMPUTER SCIENCE UNDERGRADUATE | FULL-STACK DEVELOPER

Final-semester IT undergraduate completing the **Higher Diploma in Information Technology** at SLIIT City Uni, with planned progression into the **BSc (Hons) Computer Science** top-up with the University of Bedfordshire. I build production-oriented systems across **full-stack web**, **native Android (Java)**, **secure desktop applications (Electron)**, and **advanced data pipelines**. My expansive technical portfolio ranges from enterprise restaurant management and government Customs portals, to Neo4j-driven fraud detection engines and offline-first local LLM AI assistants.

## INDUSTRIAL EXPERIENCE

### Full Stack Software Engineering Intern – RMU

Sri Lanka Customs

**CUSTOMS HEADQUARTERS, COLOMBO | DEC 1, 2025 – JAN 31, 2026 | FULL-TIME WEEKDAYS (MON-FRI)**

Worked with senior developers, engineers, and IT industry professionals on the Authorized Economic Operator (AEO) web portal and internal RMU software initiatives.

- Served as a major implementation contributor for the AEO portal, working mainly on the Lochana branch before it was merged into main.
- Contributed 39 visible Git commits and helped bring the portal to a production-ready release state under senior developer supervision.
- Worked across frontend UI, CMS-driven rendering, responsive design, animations, SEO metadata, and 3D visualization features.
- Also contributed to RMDS development, LocAI SyndicateHunter, RMS Simulator, Tariff Management Software, and customs-related web system improvements.

## ENGINEERING PROJECTS

### Ammehula Restaurant & Cafe Integrated Management System

Next.js • Express • PostgreSQL

Enterprise-grade restaurant management platform for Ammehula Cafe, Negombo, delivered as Lead Full Stack Software Engineer for a 2-year real-client university PPA project.

- Led full-stack architecture across customer ordering, admin operations, POS, KDS, delivery logistics, CRM, menu engineering, and smart inventory ERP.
- Designed a 40-model PostgreSQL and Prisma schema with JWT/RBAC security, audit logs, inventory, procurement, POS, delivery, loyalty, reservations, campaigns, and feedback workflows.
- Built real-time workflows with Socket.io, GIS delivery validation with Leaflet/Turf.js, automated procurement PDFs, Docker Compose deployment, and a 39-spec Cypress E2E suite.

### LocAI SyndicateHunter (Customs Fraud Risk Engine)

Python • Neo4j • Streamlit

Next-generation intelligence tool utilizing Graph Theory to detect organized fraud rings within import data.

- Engineered an automated Risk Analysis Engine via Cypher to map shared entities (TINs, Addresses, Phones).
- Built a high-performance ETL pipeline using Polars to ingest millions of records into a Neo4j Knowledge Graph.
- Developed an interactive Streamlit Command Center featuring real-time, physics-based 3D graph visualizations.

TECH SKILLS

FRONTEND

- HTML / CSS
- JavaScript
- Next.js & React
- TypeScript
- Vite
- Tailwind CSS
- Shadcn UI
- React Query
- Three.js
- Lottie

BACKEND & DB

- Node.js & Express
- Python & Java
- REST & WebSockets
- PostgreSQL & MySQL
- Neo4j
- Prisma ORM
- Firebase
- JWT / RBAC

DATA & ARCHITECTURE

- Polars & Pandas
- Streamlit
- Graph Theory
- Docker
- Git / GitHub
- Leaflet / Turf.js

SPECIALIZED TOOLS

- Electron.js
- Ollama (Local LLMs)
- Cypress / Playwright
- PDFKit / jsPDF
- Android Studio

PROFESSIONAL STRENGTHS

- Takes ownership from requirements to deployment-ready delivery.
- Architects diverse systems, from mobile/desktop apps to massive data pipelines.
- Comfortable learning new tools quickly and applying them in real projects.
- Able to work with senior engineers, real clients, and structured feedback.

LOC-AI Dual-Engine AI Assistant

Electron • Local LLM • OCR

- Production-ready, cross-platform desktop AI coding assistant balancing cloud power with 100% offline privacy.
- Engineered a secure Electron.js architecture with isolated IPC bridging and real-time LLM HTTP streaming.
  - Integrated a custom extraction engine featuring native Optical Character Recognition (OCR) and PDF parsing.
  - Supported dual inference: Groq API for rapid cloud reasoning and Ollama for highly sensitive local operations.

Sri Lanka Customs AEO Portal

Next.js • React • Payload CMS

- Modern Authorized Economic Operator web portal for Sri Lanka Customs, finalized for production on the main branch.
- Built responsive CMS-driven sections for hero content, benefits, process flows, tiers, and articles.
  - Developed the 3D Global Reach visualization and improved animations, SEO metadata, and mobile UX.

Additional Projects Portfolio

View All Work

Explore my complete portfolio of complex software engineering projects, technical documentation, and full source code at [lochanachamod.github.io](https://lochanachamod.github.io) and [github.com/lochanachamod](https://github.com/lochanachamod).



CORE COMPETENCIES

- ✓

Full-Stack Architecture
- ✓

Native Mobile & Desktop Arch.
- ✓

Real-Time Systems
- ✓

Graph Theory & ETL Pipelines
- ✓

Database Design & ORM
- ✓

Advanced Client-Side State
- ✓

DevOps & Containerization
- ✓

E2E Testing & QA

